PROJECT PROPOSAL

Improving Debt Consolidation Program Completion Rates

Client: Credit Canada | Prepared by: Team Broadview Analytics | March 2026

EXECUTIVE SUMMARY

Credit Canada faces a critical challenge in its Debt Consolidation Program (DCP): only approximately 50% of enrolled clients successfully complete their plans, despite average durations of 4–5 years and strong creditor support via fair-share agreements. With 2,200 annual enrollments and a dataset of 3,500 completed DCPs containing anonymized demographic, financial, and geographic variables, there is significant untapped potential to identify the drivers of success and design targeted interventions.

Team Broadview proposes a focused 10-week analytics engagement combining supervised machine learning, behavioral segmentation, and SHAP-based interpretability to deliver a risk-scoring framework and actionable recommendations to be used to increase completion rates. This aligns directly with Credit Canada's mission: helping more Canadians get out of debt and back into life.

The Business Problem

Non-profit DCPs like Credit Canada's offer powerful relief: creditors reduce or eliminate interest, consolidating unsecured debts into one affordable monthly payment with a defined timeline. Yet the 50% completion rate signals substantial attrition. Dropouts likely resume facing high-interest collections, damaged credit scores, and renewed financial stress, undermining both client outcomes and the "fair share" funding model that relies on completed repayments.

In 2023 alone, Credit Canada helped over 41,000 Canadians. Even a modest improvement in DCP completion rates would translate to hundreds of additional families achieving financial recovery annually. The key unknowns are:

- Which client profiles (demographic, financial, geographic) are most and least likely to succeed?
- What early intake indicators predict long-term adherence?
- Which modifiable factors (e.g., repayment burden relative to income, regional economic pressures) most strongly drive outcomes?

Without evidence-based prediction and segmentation, interventions remain broad and undifferentiated, missing opportunities for personalized support such as intensified check-ins for high-risk clients or region-specific financial coaching modules. Addressing this gap would increase lives impacted, enhance program reputation, and reinforce creditor confidence in the fair-share model.

Our Analytical Approach

We adopt a phased, transparent methodology grounded in the CRISP-DM framework, emphasizing interpretability and practicality for a non-profit environment.

Phase 1 — Data Preparation & Feature Engineering (Weeks 1–2)

- Ingest and clean the 3,500-record dataset: handle missing values, outliers, and inconsistencies using domain-informed imputation (median for continuous variables, mode for categoricals, strategy selected per missingness mechanism)

- **Engineer high-value features:**
  - Financial ratios: debt-to-income, repayment-to-disposable-income, debt composition by type, creditor concentration (proportion held by single largest creditor)
  - Geographic indicators: province/region, urban/rural classification, census-based economic vulnerability proxies
  - Behavioral proxies: debt vintage, number of creditors, payment burden ratio (monthly DCP obligation ÷ monthly income)
- Stratified 70/15/15 train/validation/test split to preserve class balance; scikit-learn pipeline construction to prevent data leakage
- Document all transformations for full auditability

Phase 2 — Exploratory & Diagnostic Analysis (Weeks 3–4)

- Descriptive statistics, distributional analysis, and visualizations across all feature groups
- Hypothesis testing: t-tests on income/debt levels by outcome; chi-square tests for categorical variables
- Correlation matrix and feature clustering to surface multicollinearity and guide selection
- Client segmentation via K-Means or hierarchical clustering to identify natural personas
- Geographic success gradient mapping to reveal regional patterns

Phase 3 — Predictive Modeling & Driver Identification (Weeks 5–7)

We will train and compare a suite of models suited to structured, tabular binary classification:

Model	Rationale	Interpretability	Priority
Logistic Regression (L1/L2)	Strong baseline; coefficients map directly to odds ratios for counsellor communication	High	Baseline
Decision Tree	Rule-based logic counsellors can follow; identifies threshold effects (e.g., income cut-offs)	High	Interpretability
Random Forest	Handles non-linearity and interactions; robust to outliers; Gini-based feature importance	Medium + SHAP	Performance
XGBoost / LightGBM	Best-in-class on structured data; captures complex feature interactions	Medium + SHAP	Primary

- Hyperparameter tuning via 5-fold stratified cross-validated grid search on the training set; recall prioritized to catch at-risk clients early
- SHAP values for global feature importance and local per-client explanations
- Explore causal inference techniques (propensity score methods) to distinguish true drivers from correlates

Phase 4 — Evaluation, Validation & Simulation (Weeks 8–9)

- Model performance: ROC-AUC (primary), F1-score, precision/recall, and calibration curves across all candidate models
- Business threshold optimization: precision-recall tradeoff plotted and aligned with Credit Canada's outreach capacity
- Fairness validation: performance parity assessed across demographic subgroups; sensitivity checks on key assumptions

- Intervention simulation: use model to estimate completion uplift from hypothetical changes
- Preliminary findings workshop with Credit Canada to incorporate counsellor observations

Phase 5 — Delivery & Recommendations (Week 10)

- Executive report: key success drivers ranked by impact, client personas with completion likelihoods, and 3–5 high-impact low-cost intervention recommendations with estimated uplift
- Risk-scoring framework: a validated intake tool assigning a DCP success probability and risk tier (Low / Medium / High) to each new enrollee
- Full code base model and documentation for future model updates and retraining

Expected Outcomes & Value

We anticipate delivering three categories of output:

Deliverable	Detail
Predictive Model	Validated binary classifier (target AUC > 0.75–0.80) enabling early risk flagging at intake; risk tier framework with counsellor action guides
Insight Report	Top predictors ranked by impact with effect sizes; quantified client personas (e.g., Segment A: 75% completion, Segment B: 35%); geographic success gradient analysis
Intervention Recommendations	3–5 high-impact, low-cost interventions with estimated completion uplift; prioritization framework by expected impact per counsellor hour

These insights could increase successful DCP completions by 10–20%, adding hundreds of annual successes, improving client financial wellness, strengthening fair-share revenue stability, and reinforcing Credit Canada's leadership in evidence-based credit counselling.

Why Team Broadview

Our team brings directly relevant expertise for this engagement:

- Full supervised ML pipeline proficiency in Python: scikit-learn, XGBoost, SHAP, pandas, matplotlib/seaborn
- Hands-on experience with classification problems on structured, real-world datasets including imbalanced classes, missing data, and fairness constraints
- Prior experience translating model outputs into clear, actionable recommendations for non-technical stakeholders
- Strong domain alignment: team members have engaged with financial wellness, debt management, and financial inclusion in prior coursework and professional contexts
- Commitment to a compliant, privacy-first approach throughout the engagement

Team Broadview's second choice project is BioVital.

Respectfully submitted,

Team Broadview Analytics

Queen's University Smith School of Business | MMAI 2026